

```
# Data handling libraries
import pandas as pd
import numpy as np

# Text preprocessing libraries
import re
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize

# TF-IDF library
from sklearn.feature_extraction.text import TfidfVectorizer

# Visualization libraries
import matplotlib.pyplot as plt
from wordcloud import WordCloud
```

```
# Download tokenizer and stopwords resources
nltk.download('punkt')
nltk.download('stopwords')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
True
```

```
# Load the Twitter US Airline Sentiment dataset
df = pd.read_csv("Tweets.csv")

# Display first 5 rows
df.head()
```

	tweet_id	airline_sentiment	airline_sentiment_confidence	negat
0	570306133677760513	neutral	1.0000	
1	570301130888122368	positive	0.3486	
2	570301083672813571	neutral	0.6837	
3	570301031407624196	negative	1.0000	
4	570300817074462722	negative	1.0000	

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
# Select only tweet text and sentiment label
df = df[['text', 'airline_sentiment']]

# Display dataset information
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 14640 entries, 0 to 14639
Data columns (total 2 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   text                   14640 non-null  object
1   airline_sentiment      14640 non-null  object
dtypes: object(2)
memory usage: 228.9+ KB
```

```
# Extract only negative sentiment tweets
negative_tweets = df[df['airline_sentiment'] == 'negative']

# Display number of negative tweets
negative_tweets.shape
```

```
(9178, 2)
```

```
# Load English stopwords
stop_words = set(stopwords.words('english'))

# Function to clean tweet text
def clean_tweet(text):

    # Remove URLs
    text = re.sub(r"http\S+|www\S+", "", text)

    # Remove Twitter mentions
    text = re.sub(r"@w+", "", text)

    # Remove hashtags
    text = re.sub(r"#w+", "", text)

    # Remove special characters and numbers
    text = re.sub(r"[^a-zA-Z\s]", "", text)

    # Convert text to lowercase
    text = text.lower()

    # Tokenize the text
    tokens = word_tokenize(text)

    # Remove stopwords
    tokens = [word for word in tokens if word not in stop_words]

    # Join tokens back into a string
    return " ".join(tokens)
```

```
# Apply cleaning function to negative tweets
negative_tweets['clean_text'] = negative_tweets['text'].apply(clean_tweet)

# Display original and cleaned tweets
negative_tweets[['text', 'clean_text']].head()
```

	text	clean_text
3	@VirginAmerica it's really aggressive to blast...	really aggressive blast obnoxious entertainmen...
4	@VirginAmerica and it's a really big bad thing...	really big bad thing
5	@VirginAmerica seriously would pay \$30 a fligh...	seriously would pay flight seats didnt playing...
..	@VirainAmerica SFO-PDX schedule is	

```
# Initialize TF-IDF vectorizer
tfidf = TfidfVectorizer(max_features=20)

# Fit and transform cleaned tweet text
```

```
tfidf_matrix = tfidf.fit_transform(negative_tweets['clean_text'])
```

```
# Convert TF-IDF matrix into a DataFrame
tfidf_df = pd.DataFrame(
    tfidf_matrix.toarray(),
    columns=tfidf.get_feature_names_out()
)

# Display first 5 rows of TF-IDF matrix
tfidf_df.head()
```

	amp	call	cancelled	cant	customer	delayed	flight	flightled	get	help
0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0
3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0

Next steps:

[Generate code with tfidf_df](#)[New interactive sheet](#)

```
# Calculate average TF-IDF score for each term
tfidf_scores = tfidf_df.mean().sort_values(ascending=False)

# Display top TF-IDF terms
tfidf_scores
```

	0
flight	0.161364
get	0.068515
cancelled	0.053937
service	0.052331
time	0.044687
hours	0.044553
help	0.043687
im	0.042881
customer	0.042700
hold	0.040962
plane	0.039804
us	0.038649
still	0.037706
delayed	0.037145
cant	0.036345
one	0.034583
amp	0.034461
hour	0.033592
call	0.032677
flightled	0.027828

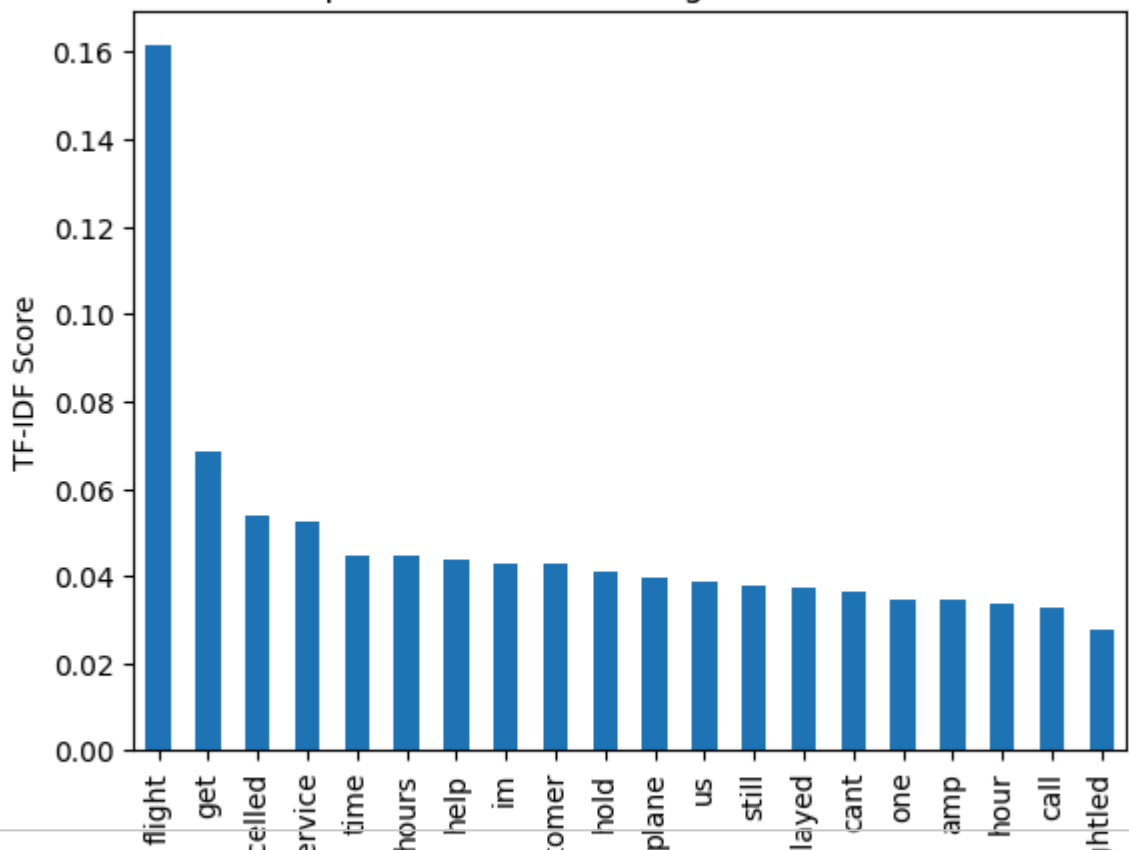
dtype: float64

```
# Plot bar chart of top TF-IDF terms
plt.figure()
tfidf_scores.plot(kind='bar')

# Add chart labels
plt.title("Top TF-IDF Terms for Negative Sentiment")
plt.xlabel("Terms")
plt.ylabel("TF-IDF Score")

# Display plot
plt.show()
```

Top TF-IDF Terms for Negative Sentiment



```
# Create word cloud from TF-IDF scores
wordcloud = WordCloud(
    width=800,
    height=400,
    background_color='white'
).generate_from_frequencies(tfidf_scores)

# Display word cloud
plt.figure(figsize=(10,5))
plt.imshow(wordcloud)
plt.axis('off')
plt.title("Word Cloud of Negative Sentiment Vocabulary")
plt.show()
```

Word Cloud of Negative Sentiment Vocabulary

